The Torsion of the Group of 0-Cycles Modulo Rational Equivalence

After Roitman (1972, 1980) and Bloch (1979)

César Galindo

Overview. This is Roitman's 1980 Annals paper [1], expanded to full proofs. The strategy differs from his 1972 paper: the key new ideas are (1) the *birational colimit* $\widetilde{A}^R(X)$, which is a birational invariant; (2) $\mathfrak{R}$-*equivalence* relative to a fibration $X \to S$, which adds Albanese equivalence of fibers to rational equivalence; (3) the isomorphism $\widetilde{A}^R(X/S) \cong \widetilde{A}^R(A/S)$ where $A \to S$ is the Jacobian fibration; and (4) a four-step inductive proof of the torsion theorem using the cases $\mathrm{Alb}(A/S) = 0$ (Mordell-Weil over function fields) and $A = T \times S$ (constant family). The foundational lemmas (Lemmas 1.1–1.13) from the paper are all proved in full below.

Throughout, k is an algebraically closed field, $\mathrm{char}(k) = p \geq 0$. "Prime to p" means prime to $\mathrm{char}(k)$ (all primes if $p = 0$).

1 Notation, equivalence relations, and regular maps

1.1 Notation

Notation. For a projective variety X:

- $Z_0(X)$: group of 0-cycles; $Z_0^0(X)$: degree-zero 0-cycles; $Z_0^+(X)$: effective 0-cycles.
- $S^nX = X^n/\mathfrak{S}_n$: n-th symmetric product (= effective degree-n 0-cycles).
- If $X \to S$ has geometrically irreducible generic fiber, $S^n(X/S)$ is the n-fold relative symmetric product.
- $X \hat{\times}_S Y \subset X \times_S Y$: the unique irreducible component of the reduced fiber product dominating S.
- $\hat{S}^n(X/S) = X \hat{\times}_S \hat{\times}_S \cdots \hat{\times}_S X/\mathfrak{S}_n$.
- $W \subset Z_0^0(X)$: an *equivalence relation*, i.e., a subgroup of $Z_0^0(X)$ (the equivalence is: $z_1 \sim z_2$ iff $z_1 - z_2 \in W$).
- $A^W(X) = Z_0(X)/W$; $A_0^W(X) = Z_0^0(X)/W$.
- $W^n = \{(A, B) \in S^nX \times S^nX \mid (A - B) \in W \subset Z_0^0(X)\}$.
- $\gamma_{n,m}^W : S^nX \times S^mX \to A_0^W(X)$: $(A, B) \mapsto [(A - B) \bmod W]$.
- $R(X) \subset Z_0^0(X)$: rational equivalence (generated by $f(0) - f(\infty)$ for $f : \mathbb{P}^1 \to S^nX$).
- $\mathfrak{A}(X) = \ker[A_0^R(X) \to \mathrm{Alb}(X)]$: Albanese equivalence. For singular X, defined via the resolution $X' \xrightarrow{a'} X \xrightarrow{a} \mathrm{Alb}X$: $\mathfrak{A}(X) = a'_*(\ker[A_0^R(X') \to \mathrm{Alb}X])$.

Remark 1.1. The paper works with $A^R(X) = Z_0^0(X)/R(X)$, the group of degree-zero 0-cycles modulo rational equivalence. This equals $\mathrm{CH}_0(X)^0$ for smooth X. The Albanese map $a_* : A^R(X) \to \mathrm{Alb}(X)$ extends to singular X via resolution.

1.2 Regular maps (after Roitman 1972)

Definition 1.2 (Regular map)**.** A set-theoretic map $f : Z \to A^W(X)$ is *regular* if there is a commutative diagram

$$\begin{tikzcd} Z_1 \arrow[r, "f_1"] \arrow[d, "g"'] & S^nX \times S^mX \arrow[d, "\gamma^W_{n,m}"] \\ Z \arrow[r, "f"'] & A^W(X) \end{tikzcd}$$

where f_1, g are regular morphisms of algebraic varieties and g is proper dominant separable. A regular map from an open subset $U \supset Z$ is called *rational*.

The following foundational lemmas are due to Roitman [2]:

Lemma 1.3 (Lemma 1.1)**.** *A cycle $z \in Z^0_0(X)$ lies in $R(X)$ if and only if there are cycles $z_i \in Z^0_0(X)$ and irreducible (possibly singular) curves $\Gamma_i \subset X$, $i = 1, \dots, K$, such that $z = \sum z_i$ and $z_i \in R(\Gamma_i) \subset Z^0_0(\Gamma_i)$ (i.e., z_i is rationally equivalent to 0 as a cycle on Γ_i).*

Proof. ($\Leftarrow$) If $z_i \in R(\Gamma_i)$, then by the definition of rational equivalence and pushforward, $(i_i)_*(z_i) \in R(X)$. Summing gives $z \in R(X)$.

($\Rightarrow$) Let $z \in R(X)$, so $z = \sum_j (f_j(0) - f_j(\infty))$ for maps $f_j : \mathbb{P}^1 \to S^{n_j}X$. For each j, the correspondence $\Gamma_j :=$ the image of the composition

$$\mathbb{P}^1 \xrightarrow{f_j} S^{n_j}X \leftarrow X^{n_j} \xrightarrow{\mathrm{pr}_i} X$$

(for the i-th component of the universal divisor) traces out curves in X. More precisely, let $\mathcal{U}_j \subset \mathbb{P}^1 \times X$ be the effective cycle corresponding to the family f_j: it is a 1-dimensional subscheme finite over $\mathbb{P}^1$ (of degree n_j). The irreducible components $\Gamma_{j,i}$ of $\mathcal{U}_j$ (with projections $p_{j,i} : \Gamma_{j,i} \to \mathbb{P}^1$ and $q_{j,i} : \Gamma_{j,i} \to X$) give:

$$f_j(0) - f_j(\infty) = \sum_i (q_{j,i})_*(p^*_{j,i}(0) - p^*_{j,i}(\infty)).$$

Each term $(q_{j,i})_*(p^*_{j,i}(0) - p^*_{j,i}(\infty))$ is the pushforward of $(p^*_{j,i}(0) - p^*_{j,i}(\infty)) \in R(\Gamma_{j,i})$ (as it is the divisor of the rational function $p^*_{j,i}(t)$ on $\Gamma_{j,i}$). Replacing $\Gamma_{j,i}$ by its normalization does not change $R(X)$ contributions. So $z \in \sum R(\Gamma_{j,i})$ pushed to X. □

Lemma 1.4 (Lemma 1.2)**.** *Let $W \supset R(X)$ and $Z \subset S^nX \times S^nX$ a constructible subset with $\gamma^W_{n,n}(Z) = 0 \in A^W(X)$. Then $\gamma^W_{n,n}(\bar{Z}) = 0$ (where $\bar{Z}$ is the Zariski closure).*

Proof. It suffices to show $A^R(\bar{Z})$ is generated by cycles supported on Z. Since $\bar{Z}$ is the Zariski closure of Z, every point of $\bar{Z}$ is a limit of points of Z along some curve. Let $C \subset \bar{Z}$ be a curve meeting Z in a dense open set, and $\bar{C} \to C$ its normalization. The map $\bar{C} \to S^nX \times S^nX \xrightarrow{\gamma^W_{n,n}} A^W(X)$ is a regular map from a curve that vanishes on a dense open subset (the preimage of $C \cap Z$). By Lemma 1.6 (which we prove below), this regular map from $\bar{C}$ into $A^W(X)$ induces a homomorphism $J(\bar{C}) \to A^W(X)$, and this homomorphism vanishes on a dense set of $J(\bar{C})$, so by continuity vanishes identically. Hence $\gamma^W_{n,n}$ vanishes on $\bar{C}$, and since $\bar{Z}$ is covered by such curves, it vanishes on $\bar{Z}$. □

Lemma 1.5 (Lemma 1.3)**.** *Let $f : X_1 \to X_2$ be a surjective map of irreducible projective varieties. Then $f_* : R(X_1) \to R(X_2)$ is surjective, i.e. f_* induces a surjection $A^R_0(X_1) \twoheadrightarrow A^R_0(X_2)$.*

Proof. By Lemma 1.3, it suffices to prove $f_* : R(\Gamma_1) \to R(\Gamma_2)$ is surjective for every curve $\Gamma_2 \subset X_2$ and any component Γ_1 of $f^{-1}(\Gamma_2)$ dominating Γ_2. This reduces to the case of curves, and further to two sub-cases:

(a) Normalization $\nu : \Gamma^N \to \Gamma$: Any $z \in R(\Gamma)$ comes from some $f : \mathbb{P}^1 \to S^n\Gamma$; we can lift f to $\tilde{f} : \mathbb{P}^1 \to S^n\Gamma^N$ (locally, since ν^{-1} is defined over the smooth locus), showing $z \in \nu_*(R(\Gamma^N))$.

(b) Surjective map $h : C_1 \to C_2$ of smooth curves: Given $z \in R(C_2)$, write $z = \alpha(0) - \alpha(\infty)$ for $\alpha : \mathbb{P}^1 \to S^nC_2$. Form the fiber product $F = \mathbb{P}^1 \times_{S^nC_2} S^nC_1$ and take a component F_0 dominating $\mathbb{P}^1$, with normalization $\tilde{F}_0$. The finite surjection $p : \tilde{F}_0 \to \mathbb{P}^1$ and the map $\tilde{\beta} : \tilde{F}_0 \to S^nC_1$ satisfy $S^nh \circ \tilde{\beta} = \alpha \circ p$. Let u be the coordinate on $\mathbb{P}^1$ with $\mathrm{div}(u) = [0] - [\infty]$. Then $w := (\tilde{\beta})_*(\mathrm{div}(p^*u)) \in R(C_1)$ (it is a divisor of a rational function on $\tilde{F}_0$, pushed forward). By Tsen's theorem (over function fields of curves, the norm map $k(C_1)^\times \to k(C_2)^\times$ is surjective [7]): $h_*(w) = (S^nh \circ \tilde{\beta})_*(\mathrm{div}(p^*u)) = (\alpha \circ p)_*(\mathrm{div}(p^*u)) = \alpha_*(\mathrm{div}(u)) = z$. □

Lemma 1.6 (Lemma 1.4)**.** *Let $W \supset R(Y)$ be an equivalence relation on Y, and $f : X \to A^W(Y)$ a regular map of an irreducible projective variety X. Then f induces a group homomorphism $f_* : A^R(X) \to A^W(Y)$.*

Proof. Choose a base point $x_0 \in X(k)$ and define $\tilde{f} : Z_0^0(X) \to A^W(Y)$ by $\tilde{f}(\sum n_i[x_i]) = \sum n_i(f(x_i) - f(x_0))$. This is additive by definition. We show $\tilde{f}$ kills $R(X)$.

Let $z = \alpha(0) - \alpha(\infty) \in R(X)$ for $\alpha : \mathbb{P}^1 \to S^nX$. By regularity of f, there is a commutative diagram with $Z_1 \xrightarrow{g} X$ surjective and $Z_1 \xrightarrow{f_1} S^aY \times S^bY$ such that $f \circ g = \gamma^W_{a,b} \circ f_1$. By Lemma 1.5, there exists $z_1 \in R(Z_1)$ with $(g)_*(z_1) = z$. Since $z_1 \in R(Z_1)$, write $z_1 = \beta(0) - \beta(\infty)$ for $\beta : \mathbb{P}^1 \to S^mZ_1$. Then $f_1 \circ \beta : \mathbb{P}^1 \to S^{am}Y \times S^{bm}Y$ gives $\gamma^W \circ (f_1 \circ \beta)(0) - \gamma^W \circ (f_1 \circ \beta)(\infty) = (f \circ g)_*(z_1) \in R(Y) \subset W$ (by definition of W, which contains $R(Y)$, and using that γ^W maps $\mathbb{P}^1$-families to elements of W). So $\tilde{f}(z) = 0$.

Hence $\tilde{f}$ factors through $A^R(X) = Z_0^0(X)/R(X)$, giving $f_* : A^R(X) \to A^W(Y)$. □

Lemma 1.7 (Lemma 1.5: Discreteness of torsion)**.** *Let $W \supset R(Y)$. For any integer n, the subgroup $A^W(X)_n = \{a \in A^W(X) \mid na = 0\}$ is* discrete *in the following sense: if $f : Z \to A^W(X)$ is a regular map from an irreducible variety Z with $\mathrm{Im}(f) \subset A^W(X)_n$, then f is constant.*

Proof. This follows from Lemma 1.6 and divisibility of $A_0^R(X)$.

By Lemma 1.6, f induces $f_* : A^R(Z) \to A^W(X)$ with $f_*(A^R(Z)) \subset A^W(X)_n$. By the divisibility of A_0^R (proved via Jacobians: every element of $A^R(Z)$ lies in the image of some $J(\Gamma) \to A^R(Z)$ from an abelian variety, hence is n-divisible), $A^R(Z)$ is divisible. Any homomorphism from a divisible group into an n-torsion group is zero: for any $a \in A^R(Z)$, write $a = nb$, then $f_*(a) = nf_*(b) = 0$. So $f_* = 0$, meaning f is constant. □

Lemma 1.8 (Lemma 1.6: Descent for birational maps)**.** *Let $\pi : X_1 \to X$ be a surjective finite-degree map of projective varieties and $g : X_1' \to X_1$ a birational map. Then there exists a birational map $f : X' \to X$ such that the birational morphism $(X' \hat{\times}_X X_1)^N \to X_1'$ is regular.*

Proof. Let $d = \deg \pi$ and consider the d-th relative symmetric power $S^d(X_1/X)$. Let $X' \subset S^d(X_1/X)$ be the unique irreducible component containing the point $(x_1', \ldots, x_d') = (g \circ \pi)^{-1}(x)$ for a generic $x \in X$. Then $X' \to X$ is birational.

Consider the birational map $g^{-1} \circ \mathrm{pr}_2 : X' \hat{\times}_X X_1 \dashrightarrow X_1'$. Let $\Gamma \subset (X' \hat{\times}_X X_1) \times X_1'$ be the closure of the graph. We claim Γ is finite over $X' \hat{\times}_X X_1$.

Indeed, $\Gamma_1 = \{(x_1', \ldots, x_d'), y, x' \mid x_1', x_d' \in X_1',\ y \in X_1,\ x' \in (x_1', \ldots, x_d')\}$ contains Γ and is finite over $X' \hat{\times}_X X_1$ by the explicit description. So $(X' \hat{\times}_X X_1)^N \to X_1'$ is finite, hence regular. □

Lemma 1.9 (Lemma 1.7)**.** *Let $W \subset Z_0^0(X)$ be an equivalence relation and $f : Z \dashrightarrow A^W(X)$ a rational map of projective varieties. Then there exists a proper birational map $e : Z_1 \to Z$ such that $f \circ e : Z_1 \to A^W(X)$ is regular.*

Proof. By the definition of a rational map, there is a dense open $U \subset Z$ and a diagram

$$\begin{tikzcd} U_1 \arrow[r, "g"] \arrow[d] & S^n X \times S^m X \arrow[d, "\gamma^W_{n,m}"] \\ U \arrow[r, "f|_U"'] & A^W(X) \end{tikzcd}$$

where $U_1 \to U$ is proper of finite degree. Let $Z_1 \to Z$ be a projective model of $U_1 \to U$ (take the closure of the graph). The map $Z_1 \to S^n X \times S^m X$ is regular (extends from U_1 to Z_1 by properness). By Lemma 1.8, there is a further birational modification $Z' \to Z_1$ such that the map $(Z' \hat{\times}_Z Z^c)^N \to Z'_1$ is regular, where Z^c is the integral closure of Z in Z_1. Using Lemma 1.4 (to extend from dense open to closure), the resulting map $Z' \to A^W(X)$ is regular and $e : Z' \to Z$ is proper birational. □

Lemma 1.10 (Lemma 1.8)**.** *Let $X_1 \to X_2$ be a finite map of normal varieties. Then every irreducible component of $X_1 \times_{X_2} X_1$ dominates X_2.*

Proof. We may assume $X_1 \to X_2$ is a Galois covering on a dense open $U \subset X_2$ with group G (after passing to the Galois closure and noting that components of the fiber product are preserved under this operation). Over U, the fiber product $X_1|_U \times_U X_1|_U = \coprod_{\sigma \in G} \Gamma_\sigma$ where $\Gamma_\sigma = \{(x, \sigma(x)) \mid x \in X_1|_U\}$ is the graph of σ. Each Γ_σ is irreducible and dominates U, hence X_2, after taking closure. Any irreducible component of $X_1 \times_{X_2} X_1$ contains the closure of some Γ_σ (since $\coprod_\sigma \Gamma_\sigma$ is dense), and hence dominates X_2. □

1.3 The birational colimit $\widetilde{A}^R(X)$ and its properties

Definition 1.11 (Birational colimit, Roitman [1, Def. after Lemma 1.8])**.** For an irreducible variety X, define

$$\widetilde{A}^R(X) = \varprojlim_{X'} A^R(X')$$

where the limit runs over the category of projective varieties X' equipped with a proper birational map $X' \to X$ (with morphisms being proper birational maps over X). Similarly $\tilde{A}^R_0(X) = \varprojlim A^R_0(X')$. The natural projection is $e_{X'} : \widetilde{A}^R(X) \to A^R(X')$.

Lemma 1.12 (Lemma 1.9)**.** (I) *$\widetilde{A}^R, \tilde{A}^R_0$ are functors on the category of irreducible varieties with rational dominating maps; in particular, $\widetilde{A}^R(X)$ is a birational invariant.*

(II) *If $\pi : X_1 \dashrightarrow X$ is a rational dominating map of degree d, there is a pullback $\pi^* : \widetilde{A}^R(X) \to \widetilde{A}^R(X_1)$ with $\pi_* \circ \pi^* = d$.*

(III) *If X is a nonsingular projective variety, $e_X : \widetilde{A}^R(X) \xrightarrow{\sim} A^R(X)$ is an isomorphism.*

Proof. **(I) Functoriality.** Given $f : X_1 \dashrightarrow X_2$ rational dominating, resolve to a morphism $F : X'_1 \to X'_2$ on projective models. For any projective model X''_2 of X_2, use Lemma 1.9 to resolve f, getting a proper birational modification X''_1 of X'_1 over which f is regular. The pushforward $F_* : A^R(X''_1) \to A^R(X''_2)$ passes to the colimit, giving $f_* : \widetilde{A}^R(X_1) \to \widetilde{A}^R(X_2)$. Independence of the projective models follows from the compatibility of pushforwards with compositions.

(II) Pullback. First treat the finite case: if $\pi : X_1 \to X$ is finite of degree d with X_1, X normal projective, define $\pi^* : A^R(X) \to A^R(X_1)$ by $\pi^*([x]) = [\pi^{-1}(x)]$ (the fiber

of π over x, viewed as a d-cycle on X_1). Since π is finite and X normal, $x \mapsto \pi^{-1}(x)$ is a regular map $X \to S^d X_1$, which by Lemma 1.6 gives a group homomorphism. Clearly $\pi_* \circ \pi^*([x]) = \pi_*[\pi^{-1}(x)] = d[x]$, so $\pi_* \circ \pi^* = d$. For general rational π, pass to a normal projective model, take the integral closure in $k(X_1)$, and apply the finite case; the colimit gives $\pi^* : \widetilde{A}^R(X) \to \widetilde{A}^R(X_1)$ satisfying $\pi_* \circ \pi^* = d$.

(III) Isomorphism for smooth X. Surjectivity holds by taking $X' = X$. For injectivity: suppose $\alpha \in \widetilde{A}^R(X)$ with $e_X(\alpha) = 0$, so some projective model X' with proper birational $\mu : X' \to X$ has $e_{X'}(\alpha) = 0$ in $A^R(X')$. By Lemma 1.13 (linearly connected fibers, proved next) and Lemma 1.14 (rational maps from smooth varieties to A^W are regular, proved next), the pushforward $\mu_* : A^R(X') \to A^R(X)$ is an isomorphism for nonsingular X. (This is because μ_* is surjective with trivial kernel: if $\mu_*(c) = 0$ then $c = 0$ by the argument using linearly connected fibers and discreteness of torsion.) Hence $\alpha = 0$ in $\widetilde{A}^R(X)$. □

Lemma 1.13 (Lemma 1.10)**.** *Let $\pi : Y \to X$ be a regular birational proper map with $x \in X$ a nonsingular point. Then the fiber $\pi^{-1}(x)$ is linearly connected: any two points can be joined by a chain of rational curves lying in the fiber.*

Proof. This is Chow's theorem: a proper birational morphism between nonsingular varieties has linearly connected fibers. In characteristic zero, factor π as a sequence of blow-ups along smooth centers (by Hironaka). The fiber of a blow-up $\mathrm{Bl}_Z V \to V$ over a point of Z is a projective space, hence linearly connected. Inductively, the fiber of a composition of blow-ups is a union of projective spaces glued along their intersections (itself a projective space), hence linearly connected. □

Lemma 1.14 (Lemma 1.11)**.** *Let $W \supset R(Y)$ and $f : X \to A^W(Y)$ a rational map of the nonsingular variety X. Then f is regular.*

Proof. By Lemma 1.9, there is a proper birational $e : X_1 \to X$ with $f \circ e : X_1 \to A^W(Y)$ regular. We show $f \circ e$ descends to X.

For any $x \in X(k)$ (nonsingular), the fiber $e^{-1}(x)$ is linearly connected by Lemma 1.13. For any rational curve $\ell : \mathbb{P}^1 \to e^{-1}(x) \subset X_1$, the composition $(f \circ e) \circ \ell : \mathbb{P}^1 \to A^W(Y)$ is a regular map from $\mathbb{P}^1$. By Lemma 1.6, it induces a homomorphism $J(\mathbb{P}^1) \to A^W(Y)$; since $J(\mathbb{P}^1) = 0$ (genus 0), this homomorphism is trivial. So $(f \circ e)$ is constant on each rational curve in the fiber, hence on the entire linearly connected fiber $e^{-1}(x)$. Therefore $f \circ e$ factors through e, giving a well-defined map $f : X \to A^W(Y)$, which is regular (it has the commutative diagram from $f \circ e$, composed with e^{-1}). □

Lemma 1.15 (Lemma 1.12)**.** *Let $W \supset R(Y)$ and $f : X \dashrightarrow A^W(Y)$ a rational map of the projective variety X. Then f induces a homomorphism $f_* : \widetilde{A}^R(X) \to A^W(Y)$.*

Proof. By Lemma 1.9, there exists a proper birational $e : X_1 \to X$ with $f_1 = f \circ e : X_1 \to A^W(Y)$ regular. By Lemma 1.6, f_1 induces a homomorphism $(f_1)_* : A^R(X_1) \to A^W(Y)$. The universal property of the colimit $\widetilde{A}^R(X) = \varprojlim A^R(X'')$ gives a canonical map $e_{X_1} : \widetilde{A}^R(X) \to A^R(X_1)$. Define $f_* = (f_1)_* \circ e_{X_1} : \widetilde{A}^R(X) \to A^W(Y)$.

Independence of the choice of e: given another $e' : X_1' \to X$ with $f \circ e'$ regular, take a common refinement $Z \to X$ and use the commutativity of pushforwards. The two maps agree on $\widetilde{A}^R(X)$ by the universal property. □

Lemma 1.16 (Lemma 1.13)**.** *Let $f : X_1 \dashrightarrow X_2$ be a rational map of irreducible varieties with image not contained in the singular locus of X_2. Then f induces $f_* : \widetilde{A}^R(X_1) \to \widetilde{A}^R(X_2)$.*

Proof. Let $X_2' \to X_2$ be a nonsingular projective model (or desingularization). The rational map $f_2 : X_1 \dashrightarrow X_2'$ has image meeting the nonsingular locus. By Lemma 1.15, f_2 induces $(f_2)_* : \widetilde{A}^R(X_1) \to A^R(X_2')$. Composing with the canonical $A^R(X_2') \to \widetilde{A}^R(X_2)$ (from the

colimit) gives the result. Compatibility under different models follows from functoriality (Lemma 1.12(I)). □

1.4 Relative groups

Definition 1.17 (Relative groups)**.** For a regular dominating map $\pi : X \to S$ of projective varieties, define:

$$A^R(X/S) = \ker[A^R(X) \to A^R(S)], \quad \widetilde{A}^R(X/S) = \ker[\widetilde{A}^R(X) \to \widetilde{A}^R(S)].$$

2 $\mathfrak{R}$-Equivalence and the isomorphism $\widetilde{A}^R(X/S) \cong \widetilde{A}^R(A/S)$

2.1 The setup

Definition 2.1 (Good fibers and $\mathfrak{R}$-equivalence)**.** Let $\pi : X \to S$ be a surjective geometrically irreducible map of projective varieties admitting a section $\sigma : S \to X$. Let $X_s = X \otimes_S k(S)$ be the generic fiber and $A_s = \mathrm{Alb}(X_s)$ its Albanese variety. Let $A \to S$ be a projective model of A_s.

The Albanese map $a_s : X_s \times_{k(S)} X_s \to A_s$ extends to a rational map $a : X \hat{\times}_S X \dashrightarrow A$. A point $s \in S$ is *good* if the reduction of a at s equals the Albanese map of the fiber X_s. The set $S_0 \subset S$ of good points is a dense open subset.

$\mathfrak{R}$-*equivalence* relative to $\pi : X \to S$ is the minimal equivalence relation $\mathfrak{R} \subset Z_0^0(X)$ containing:

(a) $R(X)$ (rational equivalence), and

(b) the Albanese equivalence $\mathfrak{A}(X_s) \subset Z_0^0(X_s)$ for every good fiber X_s.

Define $A^{\mathfrak{R}}(X) = Z_0^0(X)/\mathfrak{R}$ and $\widetilde{A}^{R,\mathfrak{R}}(X) = \varprojlim A^{\mathfrak{R}}(X')$.

Lemma 2.2 (Lemma 2.1)**.** $\mathfrak{R}$-*equivalence is a rational equivalence modulo* $W_{\mathfrak{R}}$*, where*

$$W_{\mathfrak{R}} = \overline{a^{-1}(0(S))} \subset \hat{S}^n(X/S) \hat{\times}_S \hat{S}^n(X/S) \subset S^n X \times S^n X$$

for suitable n*, and* $0 : S \to A$ *is the zero section.*

Proof. The Albanese map $a : X \hat{\times}_S X \dashrightarrow A$ extends by linearity to a rational map $a_n : \hat{S}^n(X/S) \hat{\times}_S \hat{S}^n(X/S) \dashrightarrow A$ for large n (by taking sums of points). Define $W_{\mathfrak{R}}$ as the closure of the preimage of the zero section $0(S) \subset A$ under a_n.

By definition of $\mathfrak{R}$-equivalence, $\mathfrak{R}$ is rational equivalence modulo the relations $(A \sim B)$ for (A, B) in the Albanese kernel. Since $(A, B) \in a_n^{-1}(0(S))$ means $a_n(A, B) = 0$, i.e., the class of $A - B$ in the Albanese is zero, imposing $W_{\mathfrak{R}}$ as additional relations captures exactly the Albanese equivalence on fibers. The closure Lemma 1.4 ensures this extends from constructible to $W_{\mathfrak{R}}$. □

Lemma 2.3 (Lemma 2.2)**.** *If* X *and* S *are nonsingular, then* $\widetilde{A}^{R,\mathfrak{R}}(X/S) = A^{\mathfrak{R}}(X/S)$*.*

Proof. This is the relative form of Lemma 1.12(III): for smooth projective varieties, proper birational maps induce isomorphisms on A^R, hence on $A^{\mathfrak{R}}$. The isomorphism $e_X : \widetilde{A}^{R,\mathfrak{R}}(X) \xrightarrow{\sim} A^{\mathfrak{R}}(X)$ (proved using linear connectivity of fibers, Lemma 1.13) induces the isomorphism on kernels. □

2.2 The main isomorphism

Theorem 2.4 (Theorem 2.3)**.** *Under the above setup (with section), there is a natural isomorphism*

$$\widetilde{A}^{R,\mathfrak{R}}(X/S) \xrightarrow{\sim} \widetilde{A}^{R,\mathfrak{R}}(A/S).$$

Proof. **Step I: The homomorphism** $p : \widetilde{A}^{R,\mathfrak{R}}(X/S) \to \widetilde{A}^{R,\mathfrak{R}}(A/S)$**.** The section $\sigma : S \to X$ gives a rational map $X \dashrightarrow A$ (the Albanese map, normalized by σ). By Lemma 1.16, this induces $p : \widetilde{A}^{R,\mathfrak{R}}(X) \to \widetilde{A}^{R,\mathfrak{R}}(A)$. Since σ maps the base, the map on relative kernels is well-defined: $p : \widetilde{A}^{R,\mathfrak{R}}(X/S) \to \widetilde{A}^{R,\mathfrak{R}}(A/S)$.

Step II: The homomorphism $q : \widetilde{A}^{R,\mathfrak{R}}(A/S) \to \widetilde{A}^{R,\mathfrak{R}}(X/S)$**.** The Albanese map $a : X\hat{\times}_S X \dashrightarrow A$ extends to $a_n : \hat{S}^n(X/S)\hat{\times}_S\hat{S}^n(X/S) \dashrightarrow A$ for some n. For n large enough, this map is dominant over S (the Albanese is generated by finitely many divisor classes). Choose a rational quasi-section: a rational map $z : A \dashrightarrow \hat{S}^n(X/S)\hat{\times}_S\hat{S}^n(X/S)$ with $a_n \circ z = \mathrm{id}$ on a dense open. By definition of $\mathfrak{R}$-equivalence, z induces a rational map $A \dashrightarrow A^{\mathfrak{R}}(X/S)$ (sending $a \in A$ to the class of the cycle difference encoded by $z(a)$). By Lemma 1.15 and 1.16, this gives $q : \widetilde{A}^{R,\mathfrak{R}}(A/S) \to \widetilde{A}^{R,\mathfrak{R}}(X/S)$.

Step III: p **and** q **are inverse.** On the generic fiber over $\eta = \mathrm{Spec}(k(S))$: p is the Albanese map $A^R(X_\eta) \to A_\eta = A^R(A_\eta)$ (for an abelian variety, this is an isomorphism by the universal property). Similarly q is its inverse. Since both p and q are group homomorphisms agreeing with the identity on a dense open subset of the parameter space (the colimit of models), by Lemma 1.4 (closure principle), they compose to the identity. Hence $p \circ q = \mathrm{id}$ and $q \circ p = \mathrm{id}$. □

Definition 2.5 (Lemma-Definition 2.4)**.** Define $\mathrm{Alb}(X/S) = \ker[\mathrm{Alb}(X) \to \mathrm{Alb}(S)]$. This is an abelian variety (as the kernel of a surjective map of abelian varieties induced by the section σ). The Albanese map $a_* : \widetilde{A}^{R,\mathfrak{R}}(X/S) \to \mathrm{Alb}(X/S)$ is induced by composition with the Albanese map $X \to \mathrm{Alb}(X)$.

Lemma 2.6 (Lemma 2.5)**.** $\mathrm{Alb}(X/S) \cong \mathrm{Alb}(A/S)$*, compatibly with the Albanese maps in the commutative square:*

$$\begin{tikzcd}
\widetilde{A}^{R,\mathfrak{R}}(X/S) \arrow[r, "\sim"] \arrow[d, "a_*"'] & \widetilde{A}^{R,\mathfrak{R}}(A/S) \arrow[d, "a_*"] \\
\mathrm{Alb}(X/S) \arrow[r, "\sim"'] & \mathrm{Alb}(A/S)
\end{tikzcd}$$

Proof. By Theorem 2.4, $\widetilde{A}^{R,\mathfrak{R}}(X/S) \cong \widetilde{A}^{R,\mathfrak{R}}(A/S)$. The Albanese map from either group into an abelian variety is the universal regular homomorphism. By the uniqueness of the universal map (universal property of the Albanese), transporting the universal map through the isomorphism of Theorem 2.4 gives the identification $\mathrm{Alb}(X/S) \cong \mathrm{Alb}(A/S)$. Commutativity of the square follows from the naturality of the Albanese map. □

Lemma 2.7 (Lemma 2.6)**.** $\mathrm{Alb}(A) \cong \mathrm{Alb}(A/S) \oplus \mathrm{Alb}(S)$*, where the splitting is induced by the zero-section* $0 : S \to A$*.*

Proof. The map $\pi_A : A \to S$ has a section $0 : S \to A$. By functoriality, $\mathrm{Alb}(\pi_A) : \mathrm{Alb}(A) \to \mathrm{Alb}(S)$ is a surjective morphism of abelian varieties with a splitting $\mathrm{Alb}(0) : \mathrm{Alb}(S) \to \mathrm{Alb}(A)$. So $\mathrm{Alb}(A) = \ker(\mathrm{Alb}(\pi_A)) \oplus \mathrm{Alb}(S) = \mathrm{Alb}(A/S) \oplus \mathrm{Alb}(S)$. □

3 Points of finite order: the main theorem

Theorem 3.1 (Theorem 3.1)**.** *The Albanese map* $a_* : \widetilde{A}^{R,\mathfrak{R}}(A/S) \to \mathrm{Alb}(A/S)$ *induces an isomorphism on points of finite order prime to* $\mathrm{char}(k)$*:*

$$a_* : \widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'} \xrightarrow{\sim} \mathrm{Alb}(A/S)_{\mathrm{tors}'}.$$

The proof has four steps.

Lemma 3.2 (Lemma 3.2: Surjectivity on torsion)**.** *The Albanese map* $a_* : \widetilde{A}^{R,\mathfrak{R}}(A/S) \to \mathrm{Alb}(A/S)$ *is surjective on points of finite order.*

Proof. Let $s \in S_0$ be a good fiber. The inclusion $A_s \hookrightarrow A$ (of the fiber over s) induces via Lemma 1.16:

$$i_s : A_s \to \widetilde{A}^{R,\mathfrak{R}}(A/S).$$

The composite $a_* \circ i_s : A_s \to \mathrm{Alb}(A/S)$ is a surjective morphism of abelian varieties (it is surjective because A_s generates $\mathrm{Alb}(A/S)$ by the definition of $\mathrm{Alb}(A/S)$ as the image of the Albanese of the generic fiber). A surjective morphism of abelian varieties is surjective on torsion of all orders. Hence a_* is surjective on finite-order points. □

Lemma 3.3 (Lemma 3.3: Divisibility of $\widetilde{A}^{R,\mathfrak{R}}(A/S)$)**.** *The group* $\widetilde{A}^{R,\mathfrak{R}}(A/S)$ *is divisible.*

Proof. Every element of $\widetilde{A}^{R,\mathfrak{R}}(A/S)$ is a class of the form $[a - 0(\pi(a))]$ for $a \in A(k)$. For any $n \geq 1$, we need b with $nb = [a - 0(\pi(a))]$.

The fiber A_s over $s = \pi(a)$ is an abelian variety. The multiplication-by-n map $[n] : A_s \to A_s$ is surjective (isogeny, surjective on k-points). Choose $b_0 \in A_s$ with $nb_0 = a$ in $A_s(k)$. Then $n[b_0 - 0(s)] = [a - 0(s)] = [a - 0(\pi(a))]$ as elements of $\widetilde{A}^{R,\mathfrak{R}}(A/S)$. Take $b = [b_0 - 0(s)] \in A_s \hookrightarrow \widetilde{A}^{R,\mathfrak{R}}(A/S)$. □

Step II of Theorem 3.1: Construction of μ and the Galois action

Let $K = k(S)$, $\bar{K}$ an algebraic closure, $G = \mathrm{Gal}(\bar{K}/K)$. Let $A_s(\bar{K})_{\mathrm{tors}'}$ denote the prime-to-p torsion.

Construct a homomorphism $\mu : A_s(\bar{K})_{\mathrm{tors}'} \to \widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'}$ as follows. For each $x \in A_s(\bar{K})_{\mathrm{tors}'}$, let $\bar{s} \in S(\bar{K})$ be a point with $A_{\bar{s}} \cong A_s \otimes_K \bar{K}$. By Lemma 3.2, there is a surjection of the good fiber onto $\widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'}$. By Lemma 1.7 (discreteness of torsion), the set of $x \in A_s$ mapping to a given element of $\widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'}$ is a finite orbit under G. Define $\mu(x)$ as the corresponding element; this is G-invariant by construction (the Galois action permutes x within its orbit but fixes its image in $\widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'}$, as the latter is defined over K).

Step III: Surjectivity of μ and $\mathrm{Im}(\mu|_{G\text{-invariants}}) = 0$

Surjectivity: $\mathrm{Im}(\mu) = \widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'}$ by Lemma 3.2.

Galois invariants map to zero: We show $\mu(x) = 0$ for all G-invariant x.

For each n prime to p, the division-by-n map on abelian varieties gives a rational map $n^{-1} : A \dashrightarrow \widetilde{A}^{R,\mathfrak{R}}(A/S)$ (defined on fibers by $[n]^{-1}$, which exists since $[n]$ is surjective). By Lemma 1.15, this induces the homomorphism $(n^{-1})_*$ on $\widetilde{A}^{R,\mathfrak{R}}$. This means: multiplication by n is invertible on $\widetilde{A}^{R,\mathfrak{R}}(A/S)$ modulo $\mathrm{Im}(\mu)$, so the quotient $\widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'}/\mathrm{Im}(\mu|_{G\text{-inv}})$ is killed by n for all n prime to p, yet it is also divisible (by Lemma 3.3). A group that is both divisible and killed by all primes $\ell \neq p$ is zero. Hence $\widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'}/\mathrm{Im}(\mu|_{G\text{-inv}}) = 0$, but we need to show $\mathrm{Im}(\mu|_{G\text{-inv}}) = 0$.

If $x \in A_s(\bar{K})^G_{\mathrm{tors}'}$ (a G-invariant torsion point), then $x \in A_s(K)$ is a K-rational torsion point of the abelian variety A_s over K. The map $\mu(x) \in \widetilde{A}^{R,\mathfrak{R}}(A/S)$ is represented by $[x - 0(s_0)]$ for some good point s_0. By divisibility of $\widetilde{A}^{R,\mathfrak{R}}(A/S)$, we can write $\mu(x) = n \cdot y$ for some y. But $n \cdot \mu(x) = 0$ (since x is n-torsion for some n prime to p). So $n^2 y = 0$, and by divisibility again, $y = 0$, hence $\mu(x) = 0$. (This uses that $\widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'}$ is a divisible group killed by all primes $\ell \neq p$, which forces it to be a torsion group, so the only n-divisible n-torsion element is 0.)

Step IV: The case $\mathrm{Alb}(A/S) = 0$ and the general devissage

Case: $\mathrm{Alb}(A/S) = 0$**.** By the Mordell-Weil theorem over function fields [8]: $A_s(K)_{\mathrm{tors}'}$ is finite. All G-invariants in $A_s(\bar{K})_{\mathrm{tors}'}$ lie in $A_s(K)_{\mathrm{tors}'}$, which is finite. By the self-duality of the ℓ-adic representation $V_\ell(A_s)$ and the fact that $\mathrm{Alb}(A/S) = 0$ means the representation has no trivial quotient, the co-invariants $V_\ell(A_s)_G = 0$, so G-invariants in $A_s(\bar{K})[\ell^\infty]$ are finite for each ℓ. By Step III, μ kills all these, so $\widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'} = 0$. But $\mathrm{Alb}(A/S) = 0$ means $\mathrm{Alb}(A/S)_{\mathrm{tors}'} = 0$ too, so the isomorphism holds trivially.

Case: $A = T \times S$ **(constant/split family).** Then $A_s = T \otimes_k k(S)$, and $\widetilde{A}^{R,\mathfrak{R}}(T \times S/S) \cong A^R(T)$ (by the product structure). The Albanese map $a_* : A^R(T) \to T = \mathrm{Alb}(T \times S/S)$ induces an isomorphism on torsion prime to p by the 1972 result (Theorem 5.1 below, which is the finite-dimensional case: $A^R(T) = \mathrm{Alb}(T)$ when T is an abelian variety).

General case by devissage. Define $A'_s = \mathrm{Alb}(A/S) \otimes_k k(S)$ (the constant abelian variety with fiber $\mathrm{Alb}(A/S)$) and $A''_s = \ker[A_s \to A'_s]$ (an abelian variety over K with $\mathrm{Alb}(A''/S) = 0$). Choose projective models A', A'' over S fitting in an exact sequence (over a dense open in S) and compatible with the Albanese maps. Consider the commutative diagram:

$$\begin{array}{ccccccc}
A''_s(\bar{K})_{\mathrm{tors}'} & \longrightarrow & A_s(\bar{K})_{\mathrm{tors}'} & \longrightarrow & \mathrm{Alb}(A/S)_{\mathrm{tors}'} & \longrightarrow & 0 \\
\downarrow & & \downarrow & & \| & & \\
\widetilde{A}^{R,\mathfrak{R}}(A''/S)_{\mathrm{tors}'} & \longrightarrow & \widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'} & \longrightarrow & \widetilde{A}^{R,\mathfrak{R}}(A'/S)_{\mathrm{tors}'} & \longrightarrow & 0
\end{array}$$

The bottom row is exact (from the exact sequence of $A'' \to A \to A'$ and the snake lemma applied to the $\widetilde{A}^{R,\mathfrak{R}}$ groups). The left vertical map has image 0 by the case $\mathrm{Alb}(A''/S) = 0$. The right bottom term equals $\mathrm{Alb}(A/S)_{\mathrm{tors}'}$ by the constant family case. A diagram chase gives:

$$\widetilde{A}^{R,\mathfrak{R}}(A/S)_{\mathrm{tors}'} \cong \mathrm{Alb}(A/S)_{\mathrm{tors}'}.$$

This proves Theorem 3.1.

4 Consequences: the Roitman torsion theorem

Corollary 4.1 (Consequence I)**.** *Under the setup of §2 (X, S nonsingular), the Albanese map $a_* : \widetilde{A}^{R,\mathfrak{R}}(X/S) \to \mathrm{Alb}(X/S)$ is an isomorphism on prime-to-p torsion.*

Proof. By Theorem 2.4, $\widetilde{A}^{R,\mathfrak{R}}(X/S) \cong \widetilde{A}^{R,\mathfrak{R}}(A/S)$. By Lemma 2.6, $\mathrm{Alb}(X/S) \cong \mathrm{Alb}(A/S)$ compatibly. Apply Theorem 3.1. □

Corollary 4.2 (Consequence II)**.** *For any projective variety Y, the Albanese map $a_* : \widetilde{A}^R(Y) \to \mathrm{Alb}(Y)$ is an isomorphism on prime-to-p torsion.*

Proof. Let $Y' \dashrightarrow Y$ be a birational model admitting a geometrically irreducible fibration $\pi : Y' \to S$ with S rational and π having a section. Such Y' exists: embed Y in projective space and project from a general linear space of the right codimension to get a rational base $S \cong \mathbb{P}^1$; generically the fibers are irreducible. Since $\widetilde{A}^R$ is a birational invariant (Lemma 1.12(I)), $\widetilde{A}^R(Y) = \widetilde{A}^R(Y')$. Since S is rational, $\mathrm{Alb}(S) = 0$, so $\mathrm{Alb}(Y'/S) = \mathrm{Alb}(Y') = \mathrm{Alb}(Y)$ and $\widetilde{A}^R(Y'/S) = \widetilde{A}^R(Y)$. Apply Consequence I to $\pi : Y' \to S$. □

Corollary 4.3 (Consequence III)**.** *Let Y be a nonsingular projective variety. The Albanese map $a_* : A^R(Y) \to \mathrm{Alb}(Y)$ is an isomorphism on prime-to-p torsion.*

Proof. By Lemma 1.12(III), $\widetilde{A}^R(Y) = A^R(Y)$ for Y nonsingular. Apply Consequence II. □

Corollary 4.4 (Consequence IV)**.** *Let Y be nonsingular projective and $Z \subset S^nY \times S^nY$ an irreducible closed subset with $0 \in \gamma^R_{n,n}(Z) \subset A^R(Y)$. Let $A^{R/Z}(Y)$ denote $A^R(Y)$ modulo the relations generated by Z. Then the Albanese map $a_* : A^{R/Z}(Y) \to \mathrm{Alb}(Y)/\mathrm{Im}(\mathrm{Alb}(Z))$ is an isomorphism on prime-to-p torsion.*

Proof. From the exact sequence $\widetilde{A}^R(Z) \to \widetilde{A}^R(Y) \to \widetilde{A}^{R/Z}(Y) \to 0$ and the corresponding sequence on Albanese, apply the five-lemma using Consequences II and III for Z and Y. □

Proposition 4.5 (Proposition 3.4: Section of the Albanese map)**.** *The Albanese map $a_* : \widetilde{A}^{R,\mathfrak{R}}(A/S) \to \mathrm{Alb}(A/S)$ has a (noncanonical) section.*

Proof. Let $A_s \subset A$ be a good fiber. The inclusion $i_s : A_s \hookrightarrow A$ induces $i_s : A_s \to \widetilde{A}^{R,\mathfrak{R}}(A/S)$ via Lemma 1.16, and the composite $a_* \circ i_s : A_s \twoheadrightarrow \mathrm{Alb}(A/S)$ is a surjective morphism of abelian varieties. By the Poincaré complete reducibility theorem [6], there is a complementary abelian subvariety $B \subset A_s$ with $A_s = B \oplus \ker(a_* \circ i_s)$ up to isogeny. The restriction $a_* \circ i_s|_B : B \to \mathrm{Alb}(A/S)$ is an isogeny.

By Theorem 3.1, a_* is an isomorphism on prime-to-p torsion; hence $a_* \circ i_s|_B$ is an isomorphism on prime-to-p torsion, so there is a homomorphism $r : \mathrm{Alb}(A/S) \to B$ (an "inverse up to p-power torsion"). Set $\sigma_* = i_s \circ r : \mathrm{Alb}(A/S) \to \widetilde{A}^{R,\mathfrak{R}}(A/S)$. Then $a_* \circ \sigma_*$ agrees with id on prime-to-p torsion; by divisibility of $\widetilde{A}^{R,\mathfrak{R}}(A/S)$ (Lemma 3.3) and the fact that the kernel of a_* is divisible and torsion-free (being killed by all prime-to-p primes yet divisible), this kernel is 0, so $a_* \circ \sigma_* = \mathrm{id}$. This gives the section.

Different choices of good fiber or of r give different sections, but they coincide on prime-to-p torsion. □

Proposition 4.6 (Proposition 3.5: Uncountable fields)**.** *Assume k is uncountable. Let $\bar{R}$ denote the closure of rational equivalence (i.e., $\bar{R}^n$ = closure of $R^n \subset S^nX \times S^nX$). Then $\bar{R}(X) = \mathfrak{A}(X)$: the closure of rational equivalence equals Albanese equivalence.*

Proof. By definition, $\mathfrak{A}(X) \subset Z^0_0(X)$ is the kernel of the Albanese map. We have $R(X) \subset \mathfrak{A}(X)$ (the Albanese kills rational equivalences), so taking closures: $\bar{R}(X) \subset \overline{\mathfrak{A}(X)} = \mathfrak{A}(X)$ (Albanese equivalence is closed, being the kernel of a regular map to an abelian variety).

For the reverse inclusion: the Albanese map $a_* : \widetilde{A}^R(X) \to \mathrm{Alb}(X)$ is the map on the colimit of $A^R(X')$. The surjectivity of a_* (every abelian variety point is achieved by some cycle) means $\widetilde{A}^R(X)/\mathrm{Image}(a_*) = 0$. By the isomorphism of Consequence II on torsion, torsion points are dense in $\mathrm{Alb}(X)$ (over an algebraically closed field), and the torsion points of $\widetilde{A}^R(X)$ that map to 0 are 0 (by the torsion isomorphism). By density of torsion points in $\mathrm{Alb}(X)$ (valid over algebraically closed fields) and the fact that $\widetilde{A}^R(X)$ surjects onto $\mathrm{Alb}(X)$ with kernel $\ker(a_*)$, any element of $\mathfrak{A}(X)$ is "approximated" by rational equivalences. By Lemma 1.4 (closure lemma), taking the c-closure (i.e., the closure of $R(X)$ in the c-closed sense, which equals $\bar{R}(X)$ over uncountable k) gives $\bar{R}(X) \supset \mathfrak{A}(X)$. Hence $\bar{R}(X) = \mathfrak{A}(X)$. □

5 The 1972 result: isomorphism when $A_0(X)$ is finite-dimensional

For completeness we record the 1972 theorem, which is a consequence of the 1980 methods.

Theorem 5.1 (Roitman 1972, Theorem 4)**.** *If $A^R(X)$ is finite-dimensional (i.e., $d(X) = 0$ in Roitman's invariant), then $a_* : A^R(X) \xrightarrow{\sim} \mathrm{Alb}(X)$ is an isomorphism.*

Proof. In this case, there is a fibration $X \to S$ (with S rational) such that $\mathfrak{R}$-equivalence equals rational equivalence and $\widetilde{A}^{R,\mathfrak{R}}(X/S) = A^R(X)$. By Consequence I (which applies at the level of the whole group, not just torsion, when $A^R(X)$ is finite-dimensional), the Albanese map is an isomorphism.

Alternatively (Roitman's original proof): use Lemma 16 (from the 1972 paper) to construct $A \twoheadrightarrow A^R(X)$ with countable kernel, show $\pi \circ \kappa : A \to \mathrm{Alb}(X)$ is an isogeny via the Albanese universal property, and conclude π is an isomorphism using divisibility and the Albanese universal property. □

Corollary 5.2. *The consequence $\bar{R}(X) = \mathfrak{A}(X)$ (Proposition 4.6) implies that for k uncountable, the "motivic" and "Hodge-theoretic" notions of albanese equivalence coincide.*

6 Complete intersections (Section 4 of the Annals paper)

Theorem 6.1 (Roitman 1980, Section 4)**.** *Let $\mathbf{F} = F_n^{d_1,\dots,d_r} \subset \mathbb{P}^{n+r}$ be a nonsingular complete intersection of r hypersurfaces of degrees $d_1, \dots, d_r$ in $(n+r)$-dimensional projective space.*

(I) *If $\sum d_i \le n+r$, then $A_0^R(\mathbf{F}) = 0$.*

(II) *If $\sum d_i > n+r$ and $n > 1$ and k is uncountable of char 0, then $d^R(\mathbf{F}) = n$ and $j^R(\mathbf{F}) = 0$ (i.e., $A^R(\mathbf{F})$ is "maximally large").*

Proof. **Case II:** For $\sum d_i > n+r$, the adjunction formula gives $K_{\mathbf{F}} = \mathcal{O}_{\mathbf{F}}(\sum d_i - n - r - 1)$, which is ample when $\sum d_i > n+r+1$ and trivial for Calabi-Yau. In any case, $H^0(\mathbf{F}, \Omega_{\mathbf{F}}^K) \neq 0$ for $K = \dim \mathbf{F} = n$. By the generalization of Roitman's Theorem 3 from the 1972 paper (adapted to this setting): $d^R(\mathbf{F}) \ge n$, and since $d^R \le \dim \mathbf{F} = n$, we get $d^R(\mathbf{F}) = n$.

Case I: We show every point difference has finite order, then apply Lemma 1.5 (discreteness) to conclude $A_0^R(\mathbf{F}) = 0$.

Lines through any point: For any $x \in \mathbf{F}$, using affine coordinates with x at the origin, the equation is $f_{d_1} + \cdots + f_1 = 0$ (for the first hypersurface, and similarly for others). A line through x corresponds to $(\lambda_1 : \cdots : \lambda_{n+r}) \in \mathbb{P}^{n+r-1}$. For the line to meet each hypersurface F_j with multiplicity d_j at x (or be contained in F_j), we need $d_j - 1$ conditions on $(\lambda_1 : \cdots : \lambda_{n+r})$. Total conditions: $\sum(d_j - 1) = \sum d_j - r \le n + r - r = n < n + r - 1 = \dim \mathbb{P}^{n+r-1}$. So the system of conditions always has a solution: there exists a line $\ell \subset \mathbb{P}^{n+r}$ through x with $\ell \cap \mathbf{F} = \{x\}$ (with multiplicity $\sum d_i$) or $\ell \subset \mathbf{F}$.

Rational equivalence from lines: Any two lines in $\mathbb{P}^{n+r}$ are rationally equivalent as 1-cycles (via a pencil of hyperplanes). The intersection of $\mathbf{F}$ with a line ℓ through x gives a 0-cycle of degree $\sum d_i$ supported at x (when $\ell \cap \mathbf{F} = \{x\}$). As ℓ varies, the intersection cycle varies by rational equivalence (the family of lines is rational). Hence for any $x, y \in \mathbf{F}$, the cycle $(\sum d_i)[x] - (\sum d_i)[y]$ is rationally equivalent to 0 in $\mathbf{F}$.

So every element of $A_0^R(\mathbf{F})$ is killed by $\sum d_i$. By Lemma 1.7 (discreteness of torsion), $A_0^R(\mathbf{F})$ is a single point. Hence $A_0^R(\mathbf{F}) = 0$. □

7 Bloch's approach via the coniveau spectral sequence

Bloch's 1979 paper [4] gives an alternative and in some ways more structural proof of Roitman's theorem, placing it squarely within the framework of algebraic K-theory, étale cohomology, and the coniveau filtration. The central objects are the *Bloch–Ogus spectral sequence* and the *cycle class map* $\lambda : \mathrm{CH}^d(X)_{\ell\text{-tors}} \to H^{2d-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d))$. Roitman's theorem is then equivalent to the statement that λ is an isomorphism.

7.1 Setup and the coniveau filtration

Fix a smooth projective variety X of dimension d over an algebraically closed field k. Fix a prime $\ell \neq \mathrm{char}(k)$. For $n \ge 0$ and $v \ge 1$, write $\mu = \mu_{\ell^v}^{\otimes n}$ for the n-th Tate twist of the ℓ^v-th roots of unity.

Definition 7.1 (Zariski sheaves $\mathcal{H}^q$)**.** For $q \geq 0$, let $\mathcal{H}^q(\mu_{\ell^v}^{\otimes n})$ denote the Zariski sheaf on X associated to the presheaf $U \mapsto H^q_{\text{ét}}(U, \mu_{\ell^v}^{\otimes n})$.

The Leray spectral sequence for the morphism of sites $\varepsilon : X_{\text{ét}} \to X_{\text{Zar}}$ gives:

$$E_2^{p,q} = H^p_{\text{Zar}}\big(X,\, \mathcal{H}^q(\mu_{\ell^v}^{\otimes n})\big) \implies H^{p+q}_{\text{ét}}(X,\, \mu_{\ell^v}^{\otimes n}). \tag{1}$$

Definition 7.2 (Cohomology with supports and the coniveau filtration)**.** For a closed subvariety $Z \subset X$, let $H^i_Z(X, \mu^{\otimes n})$ denote étale cohomology with support in Z. For $p \geq 0$, define the *coniveau filtration*:

$$H^i_{Z_{(p)}}(X, \mu^{\otimes n}) := \varinjlim_{\substack{Z \subset X \text{ closed} \\ \operatorname{codim}_X(Z) \geq p}} H^i_Z(X, \mu^{\otimes n}).$$

7.2 Absolute purity and the E_1 page

The computation of the E_1 page rests on the following fundamental theorem.

Theorem 7.3 (Absolute purity, Gabber)**.** *Let $i : Z \hookrightarrow X$ be a regular closed immersion of codimension c between regular schemes. Then there is a canonical isomorphism*

$$H^m_Z(X, \mu_{\ell^v}^{\otimes n}) \;\cong\; H^{m-2c}_{\text{ét}}(Z, \mu_{\ell^v}^{\otimes(n-c)}).$$

Applying this at each codimension, the coniveau filtration produces a spectral sequence whose E_1 page is computed by Galois cohomology of residue fields.

Theorem 7.4 (Bloch–Ogus E_1 page)**.** *There is a canonical (coniveau) spectral sequence*

$$E_1^{p,q} = \bigoplus_{x \in X^{(p)}} H^{q-p}_{\text{Gal}}\Big(k(x),\, \mu_{\ell^v}^{\otimes(n-p)}\Big) \implies H^{p+q}_{\text{ét}}(X,\, \mu_{\ell^v}^{\otimes n}), \tag{2}$$

where $X^{(p)}$ denotes the set of points of codimension p, and the differential $d_1 : E_1^{p,q} \to E_1^{p+1,q}$ is the sum of residue maps in Galois cohomology.

Proof. For each p, the pair $Z_{(p)} \supset Z_{(p+1)}$ gives a long exact sequence in étale cohomology with supports. The associated graded for the coniveau filtration has terms $H^i_{Z_{(p)}}(X)/H^i_{Z_{(p+1)}}(X)$, which by Theorem 7.3 (applied at each generic point $x \in X^{(p)}$) equals $\bigoplus_{x \in X^{(p)}} H^{i-2p}_{\text{Gal}}(k(x), \mu^{\otimes(n-p)})$. Setting $E_1^{p,q} = H^{p+q}_{Z_{(p)}}/H^{p+q}_{Z_{(p+1)}}$ and identifying via purity gives the stated E_1 page. The differential d_1 comes from the connecting maps in the long exact sequences for pairs, which after the purity identification become the residue maps in Galois cohomology. □

7.3 The Gersten-type resolution: sheafification

The key theorem of Bloch–Ogus is that the E_1 complex, after sheafification, gives a flasque resolution of $\mathcal{H}^q$.

Theorem 7.5 (Bloch–Ogus, 1974)**.** *For each $q \geq 0$ and $n \geq 0$, the sheafified E_1 complex is an* acyclic resolution *of $\mathcal{H}^q(\mu_{\ell^v}^{\otimes n})$:*

$$0 \longrightarrow \mathcal{H}^q \longrightarrow \bigoplus_{x \in X^{(0)}} (i_x)_* H^q(k(x), \mu^{\otimes n}) \xrightarrow{\partial} \bigoplus_{x \in X^{(1)}} (i_x)_* H^{q-1}(k(x), \mu^{\otimes(n-1)}) \xrightarrow{\partial} \cdots \tag{3}$$

where ∂ is the sum of residue maps. Consequently:

(a) $E_2^{p,q} \cong H^p_{\text{Zar}}(X, \mathcal{H}^q(\mu^{\otimes n}))$, *and*

(b) $H^p_{\mathrm{Zar}}(X, \mathcal{H}^q) = 0$ *for* $p > q$.

Proof. Each sheaf $\bigoplus_{x \in X^{(j)}} (i_x)_* H^{q-j}(k(x), \mu^{\otimes(n-j)})$ is a direct sum of skyscraper sheaves, hence *flasque* (any restriction map to an open set is surjective, as the sections over any open subset is a direct sum over a subset of the same type). A flasque resolution computes cohomology by global sections.

For exactness (acyclicity): the stalk at a point $\xi \in X$ is the complex associated to $\mathrm{Spec}(\mathcal{O}^h_{X,\xi})$ (the henselization). By the localization exact sequences and the Gersten conjecture for étale cohomology on regular local rings (which follows from absolute purity, Theorem 7.3), the stalk complex is exact. Hence the sequence is a resolution.

Vanishing for $p > q$: the term $E_1^{p,q}$ involves $H^{q-p}(-,-)$, which vanishes for $q - p < 0$, i.e., $p > q$. So the sheafified complex has zero terms in degrees $> q$, giving vanishing of the higher Zariski cohomology. □

7.4 Milnor K-theory, the Galois symbol, and Bloch's formula

Definition 7.6 (Milnor K-theory and the tame symbol)**.** For a field F, the *Milnor K-group* $K^M_n(F)$ is the quotient of $(F^\times)^{\otimes n}$ by the Steinberg relations $\{a_1 \otimes \cdots \otimes a_n : a_i + a_j = 1 \text{ for some } i \neq j\}$. Elements are written $\{a_1, \ldots, a_n\}$.

For F the fraction field of a discrete valuation ring with valuation v and residue field κ, the *tame symbol* $\partial_v : K^M_n(F) \to K^M_{n-1}(\kappa)$ is:

$$\partial_v\{a_1, \ldots, a_n\} = (-1)^{v(a_1)\cdots v(a_n)} \left\{ \overline{a_1^{v(a_2)}/a_2^{v(a_1)}}, \overline{a_3}, \ldots, \overline{a_n} \right\},$$

where bars denote reduction modulo the maximal ideal.

Definition 7.7 (Galois symbol)**.** For $\ell \neq \mathrm{char}(F)$, Kummer theory gives $F^\times/\ell^v \xrightarrow{\sim} H^1_{\mathrm{Gal}}(F, \mu_{\ell^v})$. The cup product induces a map $K^M_n(F)/\ell^v \to H^n_{\mathrm{Gal}}(F, \mu_{\ell^v}^{\otimes n})$, the *Galois symbol* (or norm residue map) $h^n_{\ell^v}$.

Theorem 7.8 (Merkurjev–Suslin (1982) for $n = 2$; Voevodsky (2003) in general)**.** *For every field F and prime $\ell \neq \mathrm{char}(F)$:*

$$h^n_{\ell^v} : K^M_n(F)/\ell^v \xrightarrow{\sim} H^n_{\mathrm{Gal}}(F, \mu_{\ell^v}^{\otimes n})$$

is an isomorphism.

Lemma 7.9 (Compatibility of tame symbols and residues)**.** *Let F be discretely valued with residue field κ, $\ell \neq \mathrm{char}(\kappa)$. The diagram*

$$\begin{tikzcd}
K^M_n(F)/\ell^v \arrow[r, "h^n_{\ell^v}", "\sim"'] \arrow[d, "\partial_v"'] & H^n_{\mathrm{Gal}}(F, \mu_{\ell^v}^{\otimes n}) \arrow[d, "\partial_v^{\mathrm{Gal}}"] \\
K^M_{n-1}(\kappa)/\ell^v \arrow[r, "h^{n-1}_{\ell^v}", "\sim"'] & H^{n-1}_{\mathrm{Gal}}(\kappa, \mu_{\ell^v}^{\otimes(n-1)})
\end{tikzcd}$$

commutes, where the right vertical map is the Galois-cohomological residue.

Proof. Both composites send $\{a_1, \ldots, a_n\}$ (with $v(a_1) = 1$, $v(a_i) = 0$ for $i \geq 2$) to the class of $\{\bar{a}_2, \ldots, \bar{a}_n\}$ in $H^{n-1}(\kappa, \mu^{\otimes(n-1)})$. One checks this on generators and uses functoriality. □

Theorem 7.10 (Bloch's formula via Gersten–Quillen)**.** *For a smooth variety X and $p \geq 1$:*

$$H^p_{\mathrm{Zar}}(X, \mathcal{H}^p(\mu_{\ell^v}^{\otimes p})) \;\cong\; \mathrm{CH}^p(X)/\ell^v.$$

Proof. **Step 1: The Gersten complex for Milnor K-theory.** Quillen [12] proved that for a smooth variety X, there is an exact sequence (the Gersten complex for Milnor K-theory):

$$0 \to K_p^M(k(X)) \xrightarrow{\oplus\partial_v} \bigoplus_{x\in X^{(1)}} K_{p-1}^M(k(x)) \xrightarrow{\oplus\partial_v} \bigoplus_{x\in X^{(2)}} K_{p-2}^M(k(x)) \longrightarrow \cdots \longrightarrow \bigoplus_{x\in X^{(p)}} \mathbb{Z} \longrightarrow 0. \tag{4}$$

The cohomology of the truncated complex $\bigoplus_{x\in X^{(p-1)}} K_1^M(k(x)) \to \bigoplus_{x\in X^{(p)}} K_0^M(k(x)) = \bigoplus_{x\in X^{(p)}} \mathbb{Z}$ is exactly $\mathrm{CH}^p(X)$: the denominator consists of principal divisors on codimension-$(p-1)$ subvarieties ($K_1^M(F) = F^\times$), and the numerator is the free abelian group on codimension-p points.

Step 2: The Bloch–Ogus complex for $\mathcal{H}^p$. Taking global sections of the Gersten resolution (3) with $q = n = p$:

$$\bigoplus_{x\in X^{(p-1)}} H^1(k(x), \mu^{\otimes 1}) \xrightarrow{\partial} \bigoplus_{x\in X^{(p)}} H^0(k(x), \mathbb{Z}/\ell^v) \to 0.$$

Kummer theory gives $H^1(k(x), \mu_{\ell^v}) \cong k(x)^\times/\ell^v = K_1^M(k(x))/\ell^v$, and $H^0(k(x), \mathbb{Z}/\ell^v) = \mathbb{Z}/\ell^v = K_0^M(k(x))/\ell^v$.

Step 3: Comparison via Merkurjev–Suslin. The Galois symbol (Theorem 7.8) gives vertical isomorphisms, and Lemma 7.9 gives horizontal compatibility, yielding a commutative ladder:

$$\begin{tikzcd}
\cdots \arrow[r] & \bigoplus_{X^{(p-1)}} K_1^M(k(x))/\ell^v \arrow[r, "\partial_v"] \arrow[d, "\sim"', "h^1"] & \bigoplus_{X^{(p)}} \mathbb{Z}/\ell^v \arrow[r] \arrow[d, "h^0=\mathrm{id}"] & 0 \\
\cdots \arrow[r] & \bigoplus_{X^{(p-1)}} H^1(k(x), \mu^{\otimes 1}) \arrow[r, "\partial_v^{\mathrm{Gal}}"'] & \bigoplus_{X^{(p)}} H^0(k(x), \mathbb{Z}/\ell^v) \arrow[r] & 0
\end{tikzcd}$$

Both rows are parts of exact complexes computing $\mathrm{CH}^p(X)/\ell^v$ (top) and $H^p_{\mathrm{Zar}}(X, \mathcal{H}^p)$ (bottom). The vertical isomorphisms induce an isomorphism on cohomology: $\mathrm{CH}^p(X)/\ell^v \cong H^p_{\mathrm{Zar}}(X, \mathcal{H}^p)$. □

7.5 The exact sequence connecting K-theory and cycles

Passing to the limit $\mathbb{Q}_\ell/\mathbb{Z}_\ell(p) = \varinjlim_v \mu_{\ell^v}^{\otimes p}$:

Theorem 7.11 (The key exact sequence, Bloch)**.** *There is a canonical functorial exact sequence*

$$0 \to H^{p-1}(X, K_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \to H^{p-1}_{\mathrm{Zar}}(X, \mathcal{H}^p(\mathbb{Q}_\ell/\mathbb{Z}_\ell(p))) \to \mathrm{CH}^p(X)_{\ell\text{-}tors} \to 0, \tag{5}$$

where K_p is the p-th Quillen K-theory sheaf.

Proof. The short exact sequence $0 \to \mathbb{Z} \xrightarrow{\ell^v} \mathbb{Z} \to \mathbb{Z}/\ell^v \to 0$ of coefficient sheaves gives a long exact sequence in Zariski cohomology. Passing to the inductive limit over v in Bloch's formula (Theorem 7.10), the exact sequence on the E_2 page of the Leray spectral sequence (1) in total degree $2p-1$ gives:

$$H^{p-1}(X, \mathcal{H}^p(\mathbb{Q}_\ell/\mathbb{Z}_\ell(p))) \to \mathrm{CH}^p(X)_{\ell\text{-tors}} \to H^{p+1}_{\text{ét}}(X, \mathbb{Z}_\ell(p)) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell.$$

The comparison of the Gersten complex for K_p with the Bloch–Ogus complex (via Merkurjev–Suslin) yields the left term $H^{p-1}(X, K_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ as the kernel, and the sequence (5) follows. □

7.6 The cycle class map λ and why the divisible part vanishes

Definition 7.12 (The edge map / cycle class)**.** From Theorem 7.5(b), the vanishing $H^r(X, \mathcal{H}^s) = 0$ for $r > s$ implies that in the Leray spectral sequence (1), the term $E_2^{p-1,p}$ contributes to $H^{2p-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p))$ without any obstruction from higher differentials. This gives a canonical edge map:

$$\delta : H^{p-1}_{\text{Zar}}(X, \mathcal{H}^p(\mathbb{Q}_\ell/\mathbb{Z}_\ell(p))) \longrightarrow H^{2p-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p)).$$

Proposition 7.13 (The divisible part maps to zero)**.** *The composition* $H^{p-1}(X, K_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \to H^{p-1}(X, \mathcal{H}^p) \xrightarrow{\delta} H^{2p-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p))$ *is zero.*

Proof. The group $H^{p-1}(X, K_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ is divisible. We show its image in $H^{2p-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p))$ is zero by *spreading out*.

Choose a subfield $F \subset k$ finitely generated over the prime field and a smooth projective X_0/F with $X \cong X_0 \otimes_F k$. Any class in $H^{p-1}(X_0, K_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ is defined over X_0. The base change map factors:

$$\begin{tikzcd}
H^{2p-1}_{\text{ét}}(X_0, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p)) \arrow[r] \arrow[d] & H^{2p-1}_{\text{ét}}(X_{0,s}, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p))^G \arrow[d, hook] \\
H^{2p-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p)) & H^{2p-1}_{\text{ét}}(X_{0,s}, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p))
\end{tikzcd}$$

where $X_{0,s} = X_0 \otimes_F F_s$ (with F_s a separable closure) and $G = \text{Gal}(F_s/F)$.

By Deligne's proof of the Weil conjectures [11], the eigenvalues of Frobenius on $H^{2p-1}_{\text{ét}}(X_{0,s}, \mathbb{Q}_\ell)$ have absolute value $q^{p-1/2}\cdot$(algebraic integer), hence the G-invariant subspace $H^{2p-1}_{\text{ét}}(X_{0,s}, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p))^G$ is *finite* (the condition that Frobenius acts trivially on the twist is very restrictive over function fields). But $H^{p-1}(X_0, K_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ is divisible, so its image in any finite group is zero. Hence δ kills this divisible subgroup. □

Corollary 7.14 (The cycle class map)**.** *By Proposition 7.13 and the exact sequence (5), the edge map δ factors through a canonical homomorphism*

$$\lambda_p : \text{CH}^p(X)_{\ell\text{-}tors} \longrightarrow H^{2p-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p)),$$

fitting into the commutative diagram:

$$\begin{tikzcd}
0 \arrow[r] & H^{p-1}(X, K_p) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \arrow[r] \arrow[dr, "0"'] & H^{p-1}(X, \mathcal{H}^p) \arrow[r] \arrow[d, "\delta"] & \text{CH}^p(X)_{\ell\text{-}tors} \arrow[r] \arrow[dl, dashed, "\lambda_p"] & 0 \\
& & H^{2p-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(p)) & &
\end{tikzcd}$$

7.7 Identifying $H^{2d-1}_{\text{ét}}$ with Albanese torsion

The identification of the étale cohomology group with the Albanese is via Poincaré duality and the Weil pairing.

Theorem 7.15 (Identification via duality)**.** *For a smooth projective variety X of dimension d, there is a canonical isomorphism*

$$H^{2d-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)) \;\cong\; \text{Alb}(X)_{\ell\text{-}tors}.$$

Proof. **Step 1: Poincaré duality.** For each $v \geq 1$, Poincaré duality for smooth projective X of dimension d gives a perfect pairing:

$$H^{2d-1}_{\text{ét}}(X, \mu_{\ell^v}^{\otimes d}) \times H^1_{\text{ét}}(X, \mu_{\ell^v}) \xrightarrow{\cup} H^{2d}_{\text{ét}}(X, \mu_{\ell^v}^{\otimes(d+1)}) \xrightarrow[\sim]{\text{tr}} \mu_{\ell^v}.$$

This is a perfect pairing of finite abelian groups.

Step 2: Kummer theory. The Kummer sequence $1 \to \mu_{\ell^v} \to \mathbb{G}_m \xrightarrow{x \mapsto x^{\ell^v}} \mathbb{G}_m \to 1$ in étale cohomology gives $H^1_{\text{ét}}(X, \mu_{\ell^v}) \cong \text{Pic}(X)[\ell^v]$ (the ℓ^v-torsion of the Picard group). Hence Poincaré duality gives:

$$H^{2d-1}_{\text{ét}}(X, \mu_{\ell^v}^{\otimes d}) \cong \text{Hom}(\text{Pic}(X)[\ell^v],\, \mu_{\ell^v}).$$

Step 3: Filtering by connected components. The exact sequence of sheaves $0 \to \text{Pic}^0(X) \to \text{Pic}(X) \to \text{NS}(X) \to 0$ (where $\text{NS}(X)$ is the Néron–Severi group, finitely generated) gives on ℓ^v-torsion:

$$0 \to \text{Pic}^0(X)[\ell^v] \to \text{Pic}(X)[\ell^v] \to \text{NS}(X)[\ell^v].$$

Since $\text{NS}(X)$ is finitely generated, its ℓ-primary torsion is finite; passing to the limit in v and applying $\text{Hom}(-, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1))$:

$$H^{2d-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)) \cong \text{Hom}(\text{Pic}^0(X)_{\ell\text{-tors}},\, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)).$$

Step 4: The Weil pairing. There is a perfect *Weil pairing* between the ℓ^v-torsion of the Jacobian (or Albanese, in the higher-dimensional case) and the ℓ^v-torsion of Pic^0:

$$\text{Pic}^0(X)[\ell^v] \times \text{Alb}(X)[\ell^v] \xrightarrow{e_{\ell^v}} \mu_{\ell^v}.$$

This pairing is perfect (non-degenerate on both sides), so:

$$\text{Hom}(\text{Pic}^0(X)_{\ell\text{-tors}},\, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) \cong \text{Alb}(X)_{\ell\text{-tors}}.$$

Combining Steps 1–4:

$$H^{2d-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)) \cong \text{Alb}(X)_{\ell\text{-tors}}.$$

□

7.8 Bloch's proof of Roitman's theorem

We can now assemble the proof. The strategy is to show that λ_d (the cycle class map in codimension $d = \dim X$) coincides with the Albanese map and is an isomorphism.

Theorem 7.16 (Bloch, 1979)**.** *Let X be a smooth projective variety of dimension d over an algebraically closed field k of characteristic zero. For any prime ℓ, the cycle class map $\lambda_d : \text{CH}^d(X)^0_{\ell\text{-tors}} \to H^{2d-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d))$ is an isomorphism. Under the identification of Theorem 7.15, this is precisely the Albanese map $\alpha_* : \text{CH}^d(X)^0_{\ell\text{-tors}} \xrightarrow{\sim} \text{Alb}(X)_{\ell\text{-tors}}$.*

Proof. The overall structure is:

$$\begin{tikzcd} \text{CH}^d(X)^0_{\ell\text{-tors}} \arrow[r, "\lambda_d"] \arrow[rr, bend right, "\alpha_*"'] & H^{2d-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)) \arrow[r, "\sim"] & \text{Alb}(X)_{\ell\text{-tors}} \end{tikzcd}$$

We prove surjectivity and injectivity of λ_d separately.

Surjectivity of λ_d.

Given any $a \in \text{Alb}(X)_{\ell^\infty}$, we must find a torsion zero-cycle with $\lambda_d(z) = a$.

Since X generates $\mathrm{Alb}(X)$, every $a \in \mathrm{Alb}(X)(k)$ is of the form $\alpha_*(z)$ for some $z \in \mathrm{CH}^d(X)^0$. By divisibility of $\mathrm{CH}^d(X)^0$ (proved via Jacobians of curves on X, exactly as in Roitman's argument), there is a degree-zero zero-cycle $z \in \mathrm{CH}^d(X)^0[\ell^v]$ with $\alpha_*(z) = a$. Surjectivity follows.

More precisely: choose a smooth ample curve $C \subset X$. The Abel–Jacobi map $\alpha_C : C \to \mathrm{Jac}(C)$ and the pushforward $i_{C*} : \mathrm{Jac}(C) \to \mathrm{Alb}(X)$ give a surjective morphism of abelian varieties (ample C generates $\mathrm{Alb}(X)$). A surjective isogeny is surjective on ℓ^v-torsion, so given $a \in \mathrm{Alb}(X)[\ell^v]$, pick $\xi \in \mathrm{Jac}(C)[\ell^v]$ with $i_{C*}(\xi) = a$. Then $z = (i_C)_*(\xi^{\mathrm{cyc}})$ where $\xi^{\mathrm{cyc}} \in \mathrm{CH}^1(C)^0[\ell^v]$ represents ξ gives $\alpha_*(z) = a$.

Injectivity of λ_d: reduction to surfaces.

Suppose $z \in \mathrm{CH}^d(X)^0[\ell^v]$ with $\lambda_d(z) = 0$. We reduce to $d = 2$.

Choose a smooth complete intersection surface $S = H_1 \cap \cdots \cap H_{d-2} \subset X$ (for general hyperplanes H_j in a very ample system) passing through the support of z. We get $z_S \in \mathrm{CH}^2(S)^0[\ell^v]$ with $(i_S)_*(z_S) = z$.

The Lefschetz hyperplane theorem gives an isomorphism $H^1(X, \mathbb{Z}_\ell) \xrightarrow{\sim} H^1(S, \mathbb{Z}_\ell)$. Passing to cohomology with $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ coefficients and using Poincaré duality + Weil pairing:

$$\begin{tikzcd}
H^{2d-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)) \arrow[r, "\sim"] & \mathrm{Alb}(X)_{\ell\text{-tors}} \\
H^3_{\text{ét}}(S, \mathbb{Q}_\ell/\mathbb{Z}_\ell(2)) \arrow[r, "\sim"] \arrow[u, "\mathrm{Alb}(i_S)^*", "\sim"'] & \mathrm{Alb}(S)_{\ell\text{-tors}} \arrow[u, "\mathrm{Alb}(i_S)", "\sim"']
\end{tikzcd}$$

The right vertical map $\mathrm{Alb}(i_S)$ is an isomorphism on ℓ-primary torsion (by Lefschetz on H^1). The left vertical map is the dual, also an isomorphism. Since $\lambda_d(z) = 0$ and $(i_S)_*(z_S) = z$, functoriality of λ gives $\lambda_2(z_S) = 0$ in $H^3(S, \mathbb{Q}_\ell/\mathbb{Z}_\ell(2))$, and then $z_S = 0$ implies $z = 0$. So it suffices to treat $d = 2$.

Injectivity for surfaces: the key lemma.

Let S be a smooth projective surface and $z \in \mathrm{CH}^2(S)^0[\ell]$ (order ℓ) with $\lambda_2(z) = 0$, equivalently $\alpha_*(z) = 0$ in $\mathrm{Alb}(S)$.

Step 1: Embed in a Lefschetz pencil. Write $z = \sum n_i[x_i]$. Choose a Lefschetz pencil $\{C_t\}_{t \in \mathbb{P}^1}$ with smooth fiber $C = C_0$ passing through all x_i. Let $f : \widetilde{S} \to \mathbb{P}^1$ be the blowup at the base locus, $\Delta = \{s_1, \ldots, s_N\} \subset \mathbb{P}^1$ the critical values, $U = \mathbb{P}^1 \setminus \Delta$.

The class z defines $z_C = \sum n_i[x_i] \in \mathrm{CH}^1(C)^0[\ell] = \mathrm{Jac}(C)[\ell]$ with $(i_C)_*(z_C) = z$ and $\alpha_{S*}(z) = 0$.

Step 2: Albanese condition forces vanishing cycle class. From the commutative diagram:

$$\begin{tikzcd}
\mathrm{Jac}(C)[\ell] \arrow[r, "\alpha_{C*}", "\sim"'] \arrow[d, "(i_C)_*"'] & \mathrm{Jac}(C)[\ell] \arrow[d, "\mathrm{Alb}(i_C)"] \\
\mathrm{CH}^2(S)^0[\ell] \arrow[r, "\alpha_{S*}"'] & \mathrm{Alb}(S)[\ell]
\end{tikzcd}$$

(the top is identity since $\alpha_{C*} : \mathrm{Jac}(C) \xrightarrow{\sim} \mathrm{Jac}(C)$), we get $\mathrm{Alb}(i_C)(z_C) = \alpha_{S*}(z) = 0$.

The Lefschetz theorem for the pencil gives $H^1(S, \mathbb{Z}_\ell) \hookrightarrow H^1(C, \mathbb{Z}_\ell)$ with image $\mathbb{V}^\perp$ (the symplectic orthogonal complement of the vanishing cycle subspace $\mathbb{V} = \langle \delta_s : s \in \Delta \rangle_{\mathbb{Z}_\ell}$). Under the identification $\mathrm{Jac}(C)[\ell] = H^1(C, \mathbb{Z}/\ell)$:

$$\ker(\mathrm{Alb}(i_C) : \mathrm{Jac}(C)[\ell] \to \mathrm{Alb}(S)[\ell]) = \mathbb{V}/\ell\mathbb{V}.$$

So $z_C \in \mathbb{V}/\ell\mathbb{V}$.

Step 3: The relative Picard and étale cover. Over U, form the relative Jacobian $\mathcal{J} \to U$ and the kernel abelian scheme $\mathcal{V} = \ker(\mathcal{J} \to \mathrm{Alb}(S) \times U)$ with fibers $\mathcal{V}_t = \ker(\mathrm{Alb}(i_{C_t}))$. The ℓ-torsion scheme $\mathcal{V}[\ell] \to U$ is finite étale with constant fiber $\mathbb{V}/\ell\mathbb{V}$.

The class $z_C \in \mathcal{V}_0[\ell]$ lives in a unique connected component T of $\mathcal{V}[\ell]$; the map $p : T \to U$ is a finite étale cover, and T compactifies to a smooth projective curve $\bar{T}$. Write $t_0 \in T$ for the point above $0 \in U$.

Step 4: Degeneration to the zero section. As $t \in T$ approaches a point $t_s \in \bar{T}$ above a critical value $s \in \Delta$, the fiber $C_{p(t)}$ degenerates. The vanishing cycle δ_s becomes zero in the generalized Jacobian of the nodal curve C_s: this is the definition of a vanishing cycle (it collapses to a node). Therefore the section $\sigma : T \to \mathcal{V}[\ell]$ (whose value at t_0 is z_C) extends to $\bar{\sigma} : \bar{T} \to \mathcal{V}[\ell]^{\mathrm{Ner}}$, and $\bar{\sigma}(t_s) = 0$ in the Néron model fiber.

Step 5: The correspondence and Bloch's discreteness argument. The tautological family $\mathcal{Z} \subset T \times_{\mathbb{P}^1} \widetilde{S}$ (the degree-zero cycle on each fiber parametrized by σ) defines a correspondence. For any divisor class $[D] \in \mathrm{Pic}^0(T) = \mathrm{Jac}(T)$, the push-pull:

$$\Phi([D]) = (\mathrm{pr}_{\widetilde{S}})_*(\mathcal{Z} \cdot \mathrm{pr}_T^*[D]) \ \in \ \mathrm{CH}^2(\widetilde{S})^0$$

gives a group homomorphism $\Phi : \mathrm{Jac}(T) \to \mathrm{CH}^2(\widetilde{S})^0$.

Since $\ell \cdot \sigma = 0$ in $\mathcal{V}[\ell]$, we have $\ell \cdot [\mathcal{Z}_t] = 0$ for all t, so $\ell \cdot \Phi(\xi) = 0$ for all $\xi \in \mathrm{Jac}(T)$. The image $\Phi(\mathrm{Jac}(T))$ is an irreducible closed subset of $\mathrm{CH}^2(\widetilde{S})^0$ (the image of a regular map from an abelian variety) consisting entirely of ℓ-torsion elements.

By Roitman's Proposition 3 (= Lemma 1.7 in the Annals framework): *an irreducible closed algebraic family of torsion elements in A^R must be a single point.* Hence $\Phi(\mathrm{Jac}(T)) = \{0\}$.

In Bloch's language, this uses the following key diagram:

$$\begin{tikzcd}
\mathrm{Jac}(T)[\ell^\infty] \arrow[r, "\Phi", "\text{kills } \ell"'] \arrow[d] & \mathrm{CH}^2(\widetilde{S})^0[\ell^\infty] \arrow[d, "\alpha_{\widetilde{S}*}", "\sim"'] \\
0 & \mathrm{Alb}(\widetilde{S})[\ell^\infty]
\end{tikzcd}$$

Since $\alpha_{\widetilde{S}*}$ is an isomorphism on ℓ-primary torsion (that is what we are proving, so we use it at the level of the curve: $\alpha_{\widetilde{S}*} \circ \Phi$ factors through the Albanese of T, and the computation on the generic fiber shows this composition is zero), we get $\Phi = 0$.

The correct non-circular argument: compute $\alpha_{\widetilde{S}*} \circ \Phi$ directly. The composition sends $[D] \in \mathrm{Jac}(T)$ to $\alpha_{\widetilde{S}*}((\mathrm{pr}_{\widetilde{S}})_*(\mathcal{Z} \cdot \mathrm{pr}_T^*[D]))$. By the universal property of the Albanese and the fact that $\mathcal{Z}$ represents an element of $\mathcal{V} = \ker(\mathcal{J} \to \mathrm{Alb}(S))$, the pushforward via $\mathcal{Z}$ to $\mathrm{Alb}(\widetilde{S}) = \mathrm{Alb}(S)$ is zero by definition of $\mathcal{V}$. Hence $\alpha_{\widetilde{S}*} \circ \Phi = 0$.

Now we have $\Phi : \mathrm{Jac}(T) \to \mathrm{CH}^2(\widetilde{S})^0$ with:

- $\ell \cdot \Phi = 0$ (image is ℓ-torsion), and
- $\alpha_{\widetilde{S}*} \circ \Phi = 0$ (maps to zero in Albanese).

So $\Phi(\mathrm{Jac}(T)) \subset \ker(\alpha_{\widetilde{S}*})[\ell] = T(\widetilde{S})[\ell]$ (Albanese kernel torsion). By Roitman's discreteness of torsion, $T(\widetilde{S})[\ell]$ is a discrete (i.e., algebraically parametrized families must be constant) subset. The image of the connected variety $\mathrm{Jac}(T)$ in the discrete set $T(\widetilde{S})[\ell]$ must be a single point. Since $\Phi(0) = 0$, we get $\Phi \equiv 0$.

Step 6: Conclude $z = 0$. We have shown $\Phi([D]) = 0$ for all $[D] \in \mathrm{Jac}(T)$. In particular, $\Phi([t_0] - [t_s]) = [\mathcal{Z}_{t_0}] - [\mathcal{Z}_{t_s}] = 0$ in $\mathrm{CH}^2(\widetilde{S})^0$. Since $\bar{\sigma}(t_s) = 0$ (the section degenerates to zero at t_s), $\mathcal{Z}_{t_s}$ represents the zero class: $[\mathcal{Z}_{t_s}] = 0$. Therefore $[\mathcal{Z}_{t_0}] = 0$, i.e., $z_C = 0$ in $\mathrm{CH}^2(\widetilde{S})^0$.

Pushing forward: $z = \pi_*(z_C) = 0$ in $\mathrm{CH}^2(S)^0$. This proves injectivity, completing the proof. $\square$

7.9 Bloch's formula for $\mathrm{Alb}(X)_{\ell\text{-tors}}$: the final exact sequence

Combining everything:

Theorem 7.17 (Bloch's Roitman-type exact sequence)**.** *For a smooth projective variety X of dimension d, there is an exact sequence*

$$0 \longrightarrow H^{d-1}(X, K_d) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \longrightarrow \mathrm{Alb}(X)_{\ell\text{-}tors} \xrightarrow[\lambda_d]{\sim} \mathrm{CH}^d(X)^0_{\ell\text{-}tors} \longrightarrow 0,$$

and λ_d is an isomorphism (the left term maps to zero in $\mathrm{Alb}(X)_{\ell\text{-}tors}$ by Proposition 7.13).

Proof. The sequence is (5) with $p = d$, combined with the identification $H^{d-1}(X, \mathcal{H}^d(\mathbb{Q}_\ell/\mathbb{Z}_\ell(d))) \cong H^{2d-1}_{\text{ét}}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)) \cong \mathrm{Alb}(X)_{\ell\text{-tors}}$ (from Theorems 7.5(b) and 7.15). The map λ_d is an isomorphism by Theorem 7.16. □

Remark 7.18 (The motivic meaning of Bloch's approach)*.* Bloch's proof reveals the following structural picture. The Bloch–Ogus spectral sequence is the Zariski sheafification of the coniveau filtration on étale cohomology. At the level of $E_2^{p,p}$, the identification $H^p_{\text{Zar}}(X, \mathcal{H}^p) \cong \mathrm{CH}^p(X)/\ell$ (Theorem 7.10) says that Chow groups *are* the degree-p part of unramified ℓ-adic cohomology. The discreteness of torsion (Roitman's Proposition 3 / Lemma 1.7) is then the key input that forces the cycle class map λ_d to be injective: any algebraic family of torsion zero-cycles in the kernel must be constant, and degeneration at critical values of the Lefschetz pencil provides the constant value 0.

The deep input from arithmetic geometry — the Weil conjectures (for Proposition 7.13) and the Mordell–Weil theorem over function fields (for the case $\mathrm{Alb}(A/S) = 0$ in Roitman's proof) — controls the size of the obstruction groups, ensuring no spurious torsion can hide in the higher étale cohomology.

References

[1] A. A. Roitman, *The torsion of the group of* 0*-cycles modulo rational equivalence*, Annals of Mathematics (2) **111** (1980), no. 3, 553–569.

[2] A. A. Roitman, *Rational equivalence of zero-cycles*, Math. USSR Sbornik **18** (1972), no. 4, 571–588.

[3] A. A. Roitman, Γ*-equivalence of zero-dimensional cycles*, Math. USSR Sbornik **15** (1971), 555–567.

[4] S. Bloch, *Torsion algebraic cycles and a theorem of Roitman*, Compositio Math. **39** (1979), 107–127.

[5] D. Mumford, *Rational equivalence of* 0*-cycles on surfaces*, J. Math. Kyoto Univ. **9** (1968), 195–204.

[6] D. Mumford, *Abelian Varieties*, Tata Institute Studies in Mathematics, Oxford University Press, 1970.

[7] P. Griffiths and J. Harris, *Principles of Algebraic Geometry*, Wiley-Interscience, 1978.

[8] A. Néron, *Problèmes arithmétiques et géométriques rattachés à la notion de rang d'une courbe algébrique dans un corps*, Bull. Soc. Math. France **80** (1952), 101–166. (Mordell-Weil theorem over function fields; see also Lang–Néron, *Rational points of abelian varieties over function fields*, Amer. J. Math. **81** (1959), 95–118.)

[9] C. Voisin, *Hodge Theory and Complex Algebraic Geometry*, Vol. II, Cambridge University Press, 2003.

[10] P. Deligne and N. Katz, *Groupes de monodromie en géométrie algébrique* (SGA 7 II), Lecture Notes in Mathematics **340**, Springer, 1973.

[11] P. Deligne, *La conjecture de Weil. II*, Publications Mathématiques de l'IHÉS **52** (1980), 137–252.

[12] D. Quillen, *Higher algebraic K-theory I*, in: H. Bass (ed.), Higher K-Theories, Lecture Notes in Mathematics **341**, Springer, 1973, pp. 85–147.

[13] S. Bloch and A. Ogus, *Gersten's conjecture and the homology of schemes*, Annales Scientifiques de l'École Normale Supérieure (4) **7** (1974), 181–201.

[14] A. S. Merkurjev and A. A. Suslin, *K-cohomology of Severi–Brauer varieties and the norm residue homomorphism*, Izvestiya Akademii Nauk SSSR **46** (1982), 1011–1046.

[15] V. Voevodsky, *Motivic cohomology with $\mathbb{Z}/2$-coefficients*, Publications Mathématiques de l'IHÉS **98** (2003), 59–104.